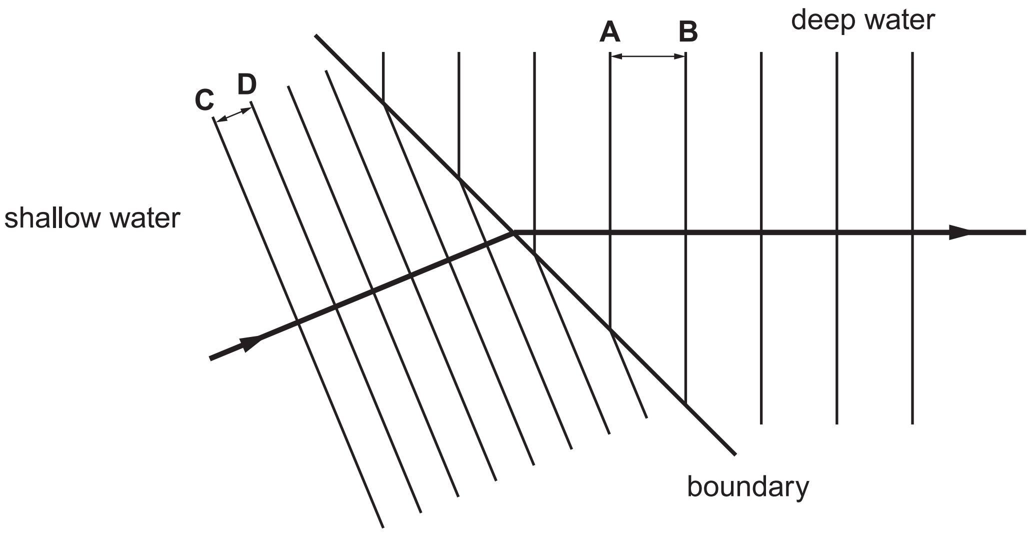
Examiner only

2. In class, a teacher demonstrates refraction using a ripple tank. The diagram below shows plane wavefronts travelling across a boundary between shallow and deep water. The frequency of the waves remains **constant** during refraction.



(a) Using a ruler, students measure the distance between wavefronts **A** and **B**. This measurement is the wavelength of the water waves in deep water. The distance between wavefronts **C** and **D** is measured to obtain their wavelength in the shallow water. The results are shown below.

	Deep water (**AB**)	Shallow water (**CD**)
Wavelength (mm)	10	5

(i) State how the measurement of wavelength could be improved. [1]

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(ii) The wavelength in the deep water is twice the wavelength in the shallow water. The teacher suggests, "the speed of the wavefronts in shallow water is double the speed of the wavefronts in the deep water." Using information provided explain if the suggestion made by the teacher is correct. [2]

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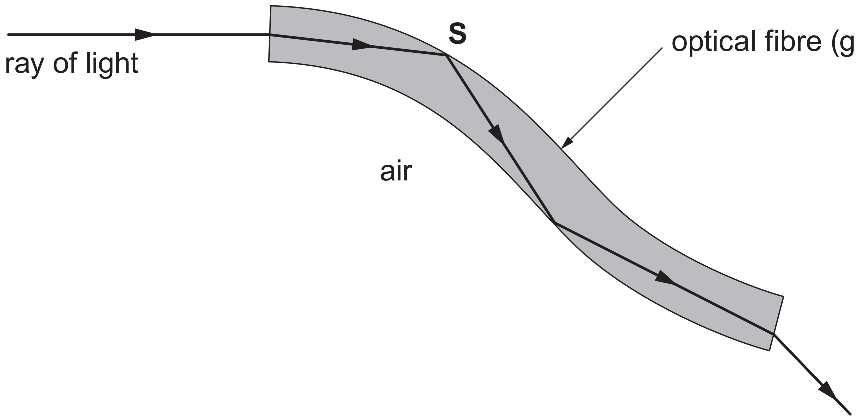
Examiner only

(b) An endoscope uses optical fibres. It can be used by doctors to produce medical images of a specific area inside a patient. A bundle of fibres is inserted into the body. Some of the fibres carry light into the body and others return the light reflected off internal surfaces. The diagram shows a ray of light passing through part of an optical fibre of an endoscope.



(i) State the name given to the change in direction of the signal at **S**. [1]

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(ii) State the **two** conditions needed for the ray of light to change direction at **S**. [2]

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(iii) Medical images can also be obtained from a computer tomography (CT) scan. This type of scan uses X-rays targeted at the patient from different positions outside the body. The information collected is processed by a computer to produce detailed 3D image segments of the patient.
Explain a disadvantage of using a CT scan to obtain medical information compared to using an endoscope. [2]

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